



FACULTY OF BUSINESS AND IT

**MBAI 5600G - Applied Integrative Analytics Capstone Project**

**Milestones 7**

**Due: July 27 & 28, and August 10 & 11**

| Milestone | Description                                      | Due Date            |
|-----------|--------------------------------------------------|---------------------|
| 1         | Capstone project selection/proposal              | May 19              |
| 2         | Literature review and data source identification | May 26              |
| 3         | Data collection & Initial EDA                    | June 9              |
| 4         | Data Preprocessing & Baseline Modeling           | June 23             |
| 5         | Advanced Modeling & Optimization                 | July 7              |
| 6         | Validation & Business Impact Assessment          | July 21             |
| 7         | Capstone project oral presentation               | July 27 & August 10 |
| 8         | Capstone project report                          | August 10           |

**Important Note on Pace and Deadlines**

Project complexity and team progress may vary throughout the term. If your group completes a milestone before its scheduled deadline, you are encouraged to begin working on the next milestone immediately rather than waiting for the due date.

**Presentation Schedule**

**July 27 (In Class)**

| Group | Start Time | End Time |
|-------|------------|----------|
| 9     | 12:00 PM   | 12:25 PM |
| 3     | 12:30 PM   | 12:55 PM |
| 10    | 1:00 PM    | 1:25 PM  |
| 20    | 1:30 PM    | 1:55 PM  |
| 18    | 2:00 PM    | 2:25 PM  |
| 11    | 2:30 PM    | 2:55 PM  |
| 2     | 3:00 PM    | 3:25 PM  |

**August 10 (In Class)**

| Group | Start Time | End Time |
|-------|------------|----------|
| 12    | 12:00 PM   | 12:25 PM |
| 14    | 12:30 PM   | 12:55 PM |
| 17    | 1:00 PM    | 1:25 PM  |
| 19    | 1:30 PM    | 1:55 PM  |
| 6     | 2:00 PM    | 2:25 PM  |
| 16    | 2:30 PM    | 2:55 PM  |
| 21    | 3:00 PM    | 3:25 PM  |

**July 28 (Online)**

**Video call link:** <https://meet.google.com/rzd-bbni-jwq>

| Group | Start Time | End Time |
|-------|------------|----------|
| 15    | 12:00 PM   | 12:25 PM |
| 13    | 12:30 PM   | 12:55 PM |
| 5     | 1:00 PM    | 1:25 PM  |

**August 11 (Online)**

**Video call link:** <https://meet.google.com/rzd-bbni-jwq>

| Group | Start Time | End Time |
|-------|------------|----------|
| 7     | 4:00 PM    | 4:25 PM  |
| 1     | 4:30 PM    | 4:55 PM  |
| 8     | 5:00 PM    | 5:25 PM  |
| 4     | 5:30 PM    | 5:55 PM  |

## **Milestone 6: Capstone Project Oral Presentation**

As the culmination of your capstone project, the final presentation is a key opportunity to demonstrate your in-depth understanding and practical application of advanced machine learning and AI concepts. Students are expected to communicate complex analytical concepts clearly, support conclusions with evidence, effectively integrate visualizations into their narrative, and maintain a strong connection between the analytical solution and the underlying business problem. The presentation should emphasize actionable insights and practical recommendations rather than focusing solely on technical details. Ensure your delivery is clear, focused, and accessible to both technical and non-technical audiences.

The **Capstone Project Presentation** is **25 minutes** in total, consisting of **up to 20 minutes for the presentation** followed by **5 minutes for questions and answers (Q&A)**. Presentations will be held on **July 27 and 28, and August 10 and 11**.

### **I. Presentation Content:**

Your presentation slides should include the following sections and adhere to the guidelines.

#### **1. Introduction (Approximately 1.5 minutes)**

- Introduce the organization, industry, or business context.
- Explain the motivation for selecting the project and why the problem is important.
- Clearly state the project objective and research question.
- Provide a brief overview of the presentation.

#### **2. Problem Definition (Approximately 2 minutes)**

- Clearly define the business problem and explain its significance.
- Translate the business problem into an analytics or AI problem.
- Identify the key stakeholders and business objectives.
- Define the project scope, assumptions, and constraints.
- Explain the criteria that will be used to evaluate project success.

#### **3. Data Description and Exploratory Data Analysis (EDA) (Approximately 3 minutes)**

- Describe the data sources, data collection process, and dataset characteristics.
- Explain the variables used in the analysis and their relevance to the business problem.
- Discuss data quality, including missing values, duplicates, and outliers.
- Present key exploratory analyses using appropriate **visualizations**.
- Highlight important patterns, relationships, and insights that informed subsequent modeling decisions.

#### **4. Methodology (Approximately 4 minutes)**

- Describe the overall analytical framework and workflow.
- Explain the data preprocessing and feature engineering techniques used.
- Justify the selection of analytical methods, machine learning algorithms, or AI techniques.
- Describe the model development process, including training, validation, and testing procedures.

- Explain the evaluation metrics used to assess model performance.
- Discuss alternative approaches considered and justify the final methodological choices.

**5. Results (Approximately 3 minutes)**

- Present the key analytical findings and model performance.
- Compare the performance of different models, if applicable.
- Interpret the results in the context of the original business problem.
- Use clear **visualizations** to communicate findings effectively.
- Explain whether the project objectives were achieved and discuss the practical significance of the results.

**6. Business Impact (Approximately 2 minutes)**

- Explain how the proposed solution creates value for the organization.
- Quantify potential business benefits whenever possible, such as cost savings, revenue growth, operational efficiency, improved customer experience, or risk reduction.
- Discuss how the findings can support managerial decision-making.
- Explain the strategic implications of implementing the proposed solution.

**7. Challenges and Limitations (Approximately 1 minute)**

- Discuss the major challenges encountered during the project.
- Explain limitations related to data availability, quality, model assumptions, scalability, or interpretability.
- Address any ethical, legal, or privacy considerations relevant to the project.
- Discuss how these limitations may affect the interpretation or generalizability of the results.

**8. Deployment and Future Work (Approximately 1.5 minutes)**

- Describe how the solution could be deployed within a real organizational environment.
- Discuss implementation considerations, including integration with existing systems and business processes.
- Explain how the model should be monitored and maintained after deployment.
- Identify opportunities for future improvements, additional data collection, or more advanced analytical approaches.

**9. Key Takeaways (Approximately 1 minute)**

- Summarize the most important findings of the project.
- Reinforce how the business problem was addressed using business analytics and AI.
- Highlight the project's key contributions to both the organization and the field of analytics.
- Conclude with a clear, concise statement that leaves the audience with the primary message of the presentation.

#### 10. Questions and Discussion (5 minutes)

Students should be prepared to:

- Justify their methodological and modeling decisions.
- Explain assumptions made throughout the project.
- Discuss alternative analytical approaches and why they were not selected.
- Respond to questions regarding implementation, scalability, ethics, and limitations with evidence-based reasoning.

#### II. Presentation Essentials

To deliver a professional and engaging presentation, students are expected to adhere to the following guidelines:

- **Practice thoroughly.** Rehearse your presentation multiple times to ensure a confident, well-paced, and seamless delivery within the allotted time.
- **Communicate clearly.** Use concise and precise language, explaining technical concepts in a way that is accessible to a diverse audience without oversimplifying the analysis.
- **Use visual aids strategically.** Incorporate high-quality charts, graphs, tables, and diagrams that reinforce key messages rather than duplicate the spoken content. Ensure all visuals are readable, appropriately labeled, and easy to interpret.
- **Engage your audience.** Maintain eye contact, speak clearly and at an appropriate pace, and use confident posture, gestures, and body language to enhance audience engagement and emphasize key points.

#### **Submission Guideline**

Please submit one **presentation file (slides)** per group through Canvas **on the day of your scheduled presentation**.